TUNIS BUSINESS SCHOOL

UNIVERSITY OF TUNIS

Cyber Security Project

2nd Deliverable

North African Hotel Pricing and Satisfaction Monitoring using Web Scraping

Participants:

Chadha Werghemi
chadha1werghemi@gmail.com

Maha Gharsalli
mahagharsalli3@gmail.com

Imen Saad
imensaad46@gmail.com

Sirine Othmene
sirineothmene1@gmail.com

Roudayna Azizi
roudayna.az505@gmail.com

IT-360 Information Assurance and Security

Pr. Manel Abdelkader

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1 Part 1: System Design, Components, and Workflow

Objective: Automatically scrape hotel listings from Booking.com across five North African countries and three seasonal date windows, extracting pricing, review scores, qualitative labels, review counts, and location data — then persist the results as a UTF-8 encoded CSV file for downstream cleaning and analysis.

1.1 User Roles

- **Developer:** Runs the scraper script, configures target countries, cities, and date scenarios, and monitors execution for connection errors or missing data.
- **Analyst:** Uses the cleaned CSV/XLSX dataset to analyse hotel pricing trends and customer satisfaction across countries and seasons.
- **System:** Selenium WebDriver + Python handles browser automation, data extraction, and CSV export — with pandas managing the cleaning pipeline and Power BI handling visualisation.

1.2 System Components

1. Scraper Script

- Uses Selenium to load and interact with Booking.com search result pages.
- Parses the DOM directly using Selenium CSS selectors.
- Extracts hotel data across all property cards per page.
- Handles connection failures with a 3-attempt retry mechanism and 15-second wait between retries.
- Encodes output in UTF-8-SIG for Excel and Power BI compatibility.

2. CSV Exporter

- Saves progress after every country–season combination to prevent data loss on connection drop.
- Uses UTF-8-SIG encoding for Excel compatibility.
- Writes all records into a structured CSV with consistent column headers.

3. Web Elements Handled

- Hotel name
- Nightly price
- Review score
- Review label
- Review count
- Location / distance from downtown

1.3 Working Flow

1. Initialise Chrome browser and construct the parameterised Booking.com search URL for the current country–season pair.
2. Attempt to load the page — retry up to 3 times with a 15-second wait if a connection error occurs.
3. Scroll to the bottom of the page three times to trigger all lazy-loaded property cards.
4. Extract all hotel fields from each property card using CSS selectors.
5. Append the record with country, city, season, and date metadata.
6. Save progress to CSV after completing each country–season combination.

7. Advance to the next country–season pair and repeat steps 1–6.
8. After all 15 combinations, quit the browser and write the final CSV file.

2 Part 2: Tools & Development Phases

2.1 Tools & Libraries

- **Python 3.10+:** Core scripting language.
- **Selenium:** Browser automation — loads JS-rendered Booking.com pages.
- **webdriver-manager:** Automatic ChromeDriver version management.
- **Pandas:** DataFrame construction, cleaning pipeline, CSV/XLSX export.
- **NumPy:** NaN handling in the cleaning pipeline.
- **re** (stdlib): Regex extraction — price, score, distance.
- **time** (stdlib): Sleep intervals between page loads and scroll operations.
- **Power BI:** Dashboard visualisation and KPI reporting.

2.2 Development Phases

- **Phase 1 – Prototype Scraper (Single City):** Initial Selenium setup targeting one city, validating CSS selectors for property cards, name, price, and review block.
- **Phase 2 – Multi-Country & Multi-Season Iteration:** Nested loop over 5 countries × 3 date scenarios, parameterised URL construction.
- **Phase 3 – Resilience & Error Handling:** Retry logic (3 attempts, 15 s wait), scroll loop with `try/except`, per-combination progress save so data is not lost on connection drop.

- **Phase 4 – Field Extraction:** Hotel name, price, review score/label/count, location (with fallback selector), all wrapped in individual `try/except` blocks.
- **Phase 5 – CSV Export with UTF-8-SIG:** Intermediate save after every country+season combination plus final export, UTF-8-SIG encoding for Excel compatibility.
- **Phase 6 – Dashboard with Power BI:** Visualisation and KPIs.

3 Part 3: Data Cleaning & Preprocessing

Overview: After scraping, the raw CSV is processed by a standalone cleaning script (`clean_hotels2.py`) consisting of 12 sequential steps. The pipeline transforms unstructured, string-typed scraped output into a fully typed, validated, analysis-ready dataset of 806 records across 11 columns.

Input
`north_africa_final_dataset.csv` (raw)
Mixed string types throughout
Duplicate rows from scroll batches
Missing values stored as string `"N/A"`

Output
`hotels_clean.xlsx` (806 records, 11 columns)
Correct numeric, float, integer, and categorical dtypes
Fully deduplicated dataset with clean index
Zero null values — invalid rows removed

3.1 Cleaning Pipeline

1. **Whitespace Strip:** All string columns passed through `str.strip()` to remove leading and trailing spaces introduced by the HTML parser.
2. **Duplicate Removal:** Exact duplicate rows dropped using `drop_duplicates()`. Index reset. Duplicates arise when the same hotel card appears in multiple scroll batches during scraping.
3. **Score Artefact Fix:** Booking.com sometimes embeds the score inside

the label field as `"Scored 8.5"`. Regex extracts the number into the score column; the label is reset to `"good"`.

4. **Score → float64:** `pd.to_numeric(errors="coerce")` converts score to float. Unparseable values receive sentinel -1.0, flagging them for removal in Step 10.
5. **Price → Int64:** Regex extracts the first digit sequence from strings like `"TND 283"` or `"TND 425"`. Cast to nullable integer `Int64`.
6. **Review Count → Int64:** Thousands separators removed, digit sequence extracted, cast to `Int64`. The raw `review_count` column is dropped and replaced by `review_count_clean`.
7. **Location → distance_km:** Regex parses both km and metre values from the location string. Metre values divided by 1000. Result stored as `distance_km_from_downtown` (float).
8. **Season Label Reformat:** Underscore-delimited labels (e.g. `May_Spring`) reformatted to parenthetical notation (e.g. `May (Spring)`) for dashboard readability.
9. **stay_nights Derivation:** Check-in and check-out strings parsed to datetime. Difference in days stored as `stay_nights` (`Int64`). Both date columns then dropped.
10. **Invalid Row Removal:** Rows removed if score equals sentinel -1.0; `review_count` is null or zero; `review_label` is empty, `"nan"`, or `"No Reviews"`. Result: 806 valid records.
11. **stars_rating Mapping:** Deterministic ordinal map applied to `review_label`: good = 1, very good = 2, excellent = 3, wonderful = 4, exceptional = 5.

Stored as `Int64`.

12. **Type Casting & Rename:** `country`, `city`, and `season` cast to pandas `Categorical` dtype, reducing memory by approximately 70%. Column `"name"` renamed to `"hotel_name"` for clarity.

3.2 Final Dataset Schema

Column	Type	Range / Values
`country`	category	Tunisia, Morocco, Algeria, Libya, Mauritania
`city`	category	5 capitals (one per country)
`season`	category	May (Spring), July (Summer), September (LowSeason)
`hotel_name`	object	359 unique hotel names
`price`	Int64	1 – 975 (local currency, numeric)
`score`	float64	1.0 – 10.0
`review_label`	object	good, very good, excellent, wonderful, exceptional
`review_count`	Int64	1 – 11,565
`distance_km_from_downtown`	float64	0.1 – 28.3 km
`stay_nights`	Int64	1 (all scenarios are 1-night searches)

4 Dashboard Creation and Data Analysis

The cleaned dataset serves as the analytical foundation for an interactive Power BI dashboard designed to monitor hotel pricing dynamics and customer satisfaction trends across North African markets. By connecting the structured XLSX output directly to Power BI, the dashboard enables stakeholders to explore the data across multiple dimensions — country, city, season, and satisfaction tier — without any additional transformation.

The dashboard is built to answer the following key business and analytical questions:

- Which country offers the most competitive average nightly price across the three seasonal windows?
- How does customer satisfaction (score and `stars_rating`) vary between May, July, and September for each market?
- What is the distribution of hotel quality tiers (Good, Very Good, Excellent, Wonderful, Exceptional) per country?
- Is there a correlation between distance from downtown and nightly price or review score?
- Which city presents the highest concentration of high-review-count hotels, indicating market maturity?
- How does pricing fluctuate between peak season (July) and low season (September) across the five capitals?

This visualisation layer transforms the raw scraped and cleaned data into actionable intelligence, directly supporting data-driven decision-making for pricing strategy, market benchmarking, and regional tourism analysis across Tunisia, Morocco, Algeria, Libya, and Mauritania.

5 Conclusion

This document outlines the design and components of the proposed web scraping solution for extracting hotel pricing and customer satisfaction across North Africa. It covers the roles of key users, system components, working flow, and the tools required, including Python, Selenium, and Power BI. The development phases — from single-city prototype to multi-country iteration with built-in fault tolerance — are clearly defined. By following this approach, the solution ensures producing a clean, analysis-ready dataset of 806 hotel records across 5 countries and 3 seasonal windows.